



Zero-shot framework for construction equipment task monitoring

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Abstract

Vision-based monitoring of construction equipment is limited in scalability due to the high resource demands of collecting and labeling large datasets across diverse environments. This study proposes a framework that employs Zero-Shot Learning (ZSL) and Multimodal Large Language Model (MLLM) to recognize construction equipment tasks from video frames without additional training data. The framework operates in two stages: (i) a zero-shot construction equipment detection stage that includes detection and tracking modules and (ii) an MLLM-based monitoring stage, utilizing the proprietary model (i.e., GPT-4o mini) to recognize tasks. Experiments showed that the framework achieved an F1-score of 82.2% for equipment detection using ZSL. A Multiple Choice Question (MCQ) dataset was constructed for evaluating MLLM, which achieved an accuracy of 79.0%. A practical case study, focusing on excavator tasks, demonstrated accurate recognition of both idle states and complex operations. These results highlight the proposed framework's potential to automate construction equipment monitoring.

1 | INTRODUCTION

The issue of low productivity in the construction industry has been a long-standing and well-known problem. To overcome these challenges and enhance productivity, digital transformation incorporating innovative technologies such as Artificial Intelligence (AI) and Internet of Things (IoT) sensors has been actively pursued (Barbosa et al., 2017; Kang et al., 2024). The digital transformation of the construction industry has been effectively introduced in areas such as the automation of productivity analysis (Baek et al., 2024; Harichandran et al., 2023), accident prevention systems (Shen et al., 2021), and structure health monitoring (Pan et al., 2023). To enable digital

transformation, collecting information from construction sites through monitoring systems is essential. These data enable site managers to accurately understand the operation of construction equipment, the progress of tasks, and the use of resources. Moreover, it serves as a fundamental step for automation and data-driven solutions for construction equipment monitoring. Since construction equipment accounts for 25%–40% of the total project cost, it is crucial to include such equipment in monitoring systems, along with workers and construction materials, as key data points (Phadatare & Charhate, 2016).

Traditionally, construction equipment monitoring was conducted through manual observation, where site managers directly recorded equipment tasks. However, as

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construction sites have become larger and more complex, manual observation has become time-consuming and prone to human error. With the advancement of AI technologies and the increasing demand for digital transformation, vision-based methods, which utilize videos captured by Closed Circuit TeleVision (CCTV) installed at construction sites or by Unmanned Aerial Vehicles (UAVs), have emerged as promising non-contact methods (Reja et al., 2022). Previous studies have shown that vision-based methods can reliably provide information on the location and operations of construction equipment using deep learning algorithms, such as Convolutional Neural Networks (CNNs; S. Jung et al., 2022, 2023). Moreover, these methods have been applied for various purposes, such as safety management (Park et al., 2024) and productivity analysis (J. Kim & Chi, 2020).

Although previous studies showed high reliability in construction equipment monitoring, the large amount of data and labeling work still remains a challenge in ensuring the performance of vision-based methods (J. Kim et al., 2024). To overcome this, several studies have explored vision-based methods using Few-Shot Learning (FSL) and Zero-Shot Learning (ZSL) (Gil & Lee, 2024; Tang et al., 2024). FSL is a method that recognizes new tasks and identifies unseen classes using only a small number of labeled example images, while ZSL can perform these tasks without any examples. Regarding FSL, J. Kim and Chi (2021) showed that FSL could achieve more than twice the object detection performance, compared to traditional supervised learning when using 20 images. Wang and El-Gohary (2024), using FSL-based object detection, achieved 88.2% precision in detecting 39 construction-related classes with only 50 example images. Although monitoring systems using FSL have effectively recognized new construction equipment with a few example images, they still require domain-specific data to ensure performance. Moreover, because FSL-based object detection requires training with extensive labeled data for relevant classes, it remains difficult to apply such methods directly in real-world construction environments.

Since ZSL does not require any example images, it can improve the applicability of the monitoring system. However, to ensure reliable ZSL performance, it is necessary to employ a pre-trained model using a large dataset. In the field of computer science, researchers have collected approximately 400 million image–text pairs (Radford et al., 2021), which were then used to train Vision-Language Models (VLMs), such as Contrastive Language-Image Pre-training (CLIP). By learning generalized knowledge from vast amounts of data, VLMs have offered a robust foundation for ZSL. Accordingly, several studies have successfully

used VLMs to recognize various elements in construction site images, including construction objects, personal protective equipment, and excavator activities (e.g., loading and digging), demonstrating the potential of ZSL and VLMs in construction monitoring applications (Gil & Lee, 2024; Liang et al., 2024).

However, vision-based methods using ZSL still have limitations in construction equipment monitoring. First, since these methods are designed to process single images, it is difficult to capture changes in construction equipment over time in videos. Second, while VLMs like CLIP are pre-trained on general image–text pairs, they lack sufficient domain-specific data from construction sites, which are still not sufficiently reflected.

To address these limitations, this study proposes a framework for monitoring construction equipment tasks using a Multimodal Large Language Model (MLLM) capable of processing multiple video frames. MLLM is designed to simultaneously interpret different modalities, such as text, images, and audio, demonstrating reliable performance in vision-language tasks that require semantic reasoning—for example, generating detailed descriptions of images and understanding scenes in images (Lin et al., 2023). Notably, MLLM's ability to handle long inputs allows it to effectively analyze temporal changes and tasks in construction equipment across multiple video frames. Additionally, MLLM has been trained on an extensive multimedia dataset containing over 5 billion image–text pairs, significantly enhancing its generalizability, which holds great potential for collecting information on complex construction equipment tasks with ZSL. The novelty of this study can be summarized as follows.

1. This study integrates vision-based methods using ZSL for detection and tracking (i.e., Florence-2 and Segment Anything Model 2, SAM2) with MLLM to recognize construction equipment tasks by processing multiple video frames.
2. An Multiple Choice Question (MCQ) dataset was constructed, covering three question types: spatial inference, task inference, and temporal event. This dataset contributes to clarifying the scope and depth of information that MLLMs can provide for construction equipment monitoring.
3. A framework is proposed that does not require any domain-specific data from construction sites by utilizing ZSL. Specifically, the proposed framework was designed to function across the entire process, including detection, tracking, and MLLM-based monitoring, without the need for additional model training.



2 | RELATED WORKS

2.1 | Construction monitoring using vision-based methods

Vision-based methods for construction equipment monitoring have become an accessible approach as cameras have become widespread on construction sites. One of the most significant applications in this area is the object detection method. Object detection methods have gained reliability and widespread use in the construction domain due to the rapid advancements in deep learning algorithms, such as You Only Look Once (YOLO; D. Kim et al., 2019), and Faster Region-based Convolutional Neural Network (R-CNN; J. Kim & Chi, 2019). They are used to retrieve localization information (i.e., bounding box) and class information of construction objects (i.e., type of construction equipment) from a single image.

The other application of vision-based methods involves activity recognition. Activity recognition methods are used to identify the activities of construction equipment from video and image. This method typically involves multiple stages: object detection, tracking, and activity recognition. First, an object detection model extracts the bounding boxes and class information of various objects from each frame. These results are then passed to a tracking model, which assigns a unique ID to each detected object and tracks their movement across consecutive frames. Finally, the tracked data are organized into a sequence of frames (e.g., 16 or 32 frames) and is input into the activity recognition model to classify the activities of construction equipment. For example, J. Kim and Chi (2019) used Faster R-CNN and tracking algorithms to obtain the bounding boxes of an excavator from each frame. They then applied a CNN and long short-term memory-based activity recognition model to classify the activities, such as digging, dumping, and hauling. Similarly, C. Chen et al. (2020) used a 3D CNN-based model to classify three activities of an excavator (i.e., digging, loading, and swinging) and employed this information to automatically calculate the earthmoving cycle. Consequently, the activity recognition method is widely used in construction monitoring systems to extract semantic information from spatiotemporal data, enabling tasks such as calculating the earthmoving cycle (Roberts & Golparvar-Fard, 2019) and analyzing workers' productive and non-productive states (Luo et al., 2019).

Despite the contributions of vision-based methods to construction monitoring systems, several limitations remain. Most studies have relied on supervised learning to train models, which require large volumes of data. The process of collecting a substantial number of images or videos and accurately labeling them consumes significant time and resources. To mitigate these challenges, FSL and ZSL approaches have been introduced in the

construction sector. For example, J. Kim and Chi (2021) applied few-shot transfer learning with only 20 images of construction equipment, achieving a mean average precision exceeding 70% in object detection. Similarly, Cui et al. (2022) employed FSL to recognize rarely occurring building facade defects, such as efflorescence and spalling with exposed rebars, using 30 images. Liang et al. (2024) employed the CLIP model within an FSL framework to classify objects temporarily used on construction sites, such as materials and tools. They computed cosine similarity scores to measure the similarity between image and text pairs. Gil and Lee (2024) detected human body poses to extract head-only images, and then applied ZSL to detect personal protective equipment. C. Chen et al. (2023) used YOLO to detect excavators in videos and employed an object tracker to capture excavator occurrences across consecutive frames. They then input 25 detected frames into a ZSL model to classify three activities (i.e., loading, digging, and idling), achieving an accuracy of 86% for excavator activity recognition. Overall, FSL and ZSL have overcome the challenges posed by data scarcity in construction sites and demonstrated reliable performance.

2.2 | MLLMs

Large Language Models (LLMs) based on Transformer architecture have rapidly advanced, integrating billions of parameters to perform a wide range of language tasks, such as machine translation, text summarization, and topic modeling. Notable examples include the Generative Pre-trained Transformer (GPT) and Bidirectional Encoder Representations from Transformers (BERT). It has been reported that increasing the size of model and dataset directly enhances the performance of LLMs (Kaplan et al., 2020). Surprisingly, while LLMs structurally share learning algorithms and model architecture with smaller models, they exhibit new capabilities such as in-context learning, which allows them to temporarily learn new tasks from prompts. For example, GPT-3 demonstrated FSL, enabling it to solve new tasks with only a few examples, a capability not observed in GPT-2 (Radford et al., 2019). Another example is ChatGPT, which has shown effective performance in dialogue-based tasks and demonstrated the ability to follow instructions without explicit task-specific examples through instruction tuning (Wei et al., 2022). This has effectively addressed existing model limitations, such as the cold-start problem, where adequate training data are unavailable. Accordingly, utilizing LLMs and generative AI models to develop human-like systems for various applications has emerged as a new research direction, showing significant potential for system automation.



Leveraging the effective performance of LLMs, MLLMs have emerged by combining the complementary strengths of LLMs and VLMs. MLLMs generally consist of three parts: a modality encoder, a pretrained LLM, and a modality interface (Yin et al., 2024). Modality encoders process inputs from various modalities—such as images, videos, and audio—by converting them into embeddings that LLMs can interpret. The pretrained LLMs then perform reasoning based on these multimodal inputs, while the modality interface integrates information obtained from different modalities. MLLMs with this architecture have demonstrated superior performance in vision-language tasks, such as generating images from text descriptions, identifying and classifying objects and details in images, and answering questions about image content. They have also been applied across various industry domains, including medicine, education, and robotics (Z. Chen et al., 2024). Owing to the applicability of MLLMs, proprietary models such as Google's Gemini, and OpenAI's ChatGPT have rapidly advanced through competition. In particular, proprietary models offer Application Programming Interfaces (APIs), allowing non-experts who may lack the computing resources to run large-parameter MLLMs to use them easily (Carolan et al., 2024). Additionally, between 2023 and 2024, API costs significantly decreased, creating an environment where proprietary models could be adopted economically. For example, the cost of GPT-4o in August 2024 was approximately one tenth that of GPT-4 in March 2023, reaching \$3 per million tokens (Nebuly, 2024).

Meanwhile, to effectively utilize in-context learning in MLLMs, prompt engineering that adjusts both text prompts and visual inputs has been actively explored (Shtedritski et al., 2023; Yong et al., 2023). Regarding the text prompt, a representative method is to provide examples or additional information and context, similar to FSL, to elicit accurate and consistent responses. For example, Yong et al. (2023) demonstrated that adding domain-specific definitions to text prompts can improve performance when detecting building defects. Regarding visual inputs, visual prompt engineering modifies image pixels to direct the attention to specific regions. This is not limited to simply cropping images but can also involve emphasizing specific elements within the image. For example, Shtedritski et al. (2023) showed that drawing a red circle around an object effectively directs attention to that region while preserving the global image context. Mani et al. (2022) demonstrated that placing a red dot within an image could resolve ambiguity in visual question answering, thereby improving the accuracy of responses. Pu, Yang, Shi, and Jin (2024) proposed a method that employs a mask segmentation model to segment regions in construction images and assign optimal alphanumeric marks to each area, thereby enhancing the visual inputs for multimodal LLMs.

2.3 | Research gaps

MLLMs have effectively addressed various vision-language tasks via ZSL and have shown promise for applications such as construction site monitoring. Accordingly, several studies have applied MLLMs to construction-site contexts to develop automated systems (H. Chen et al., 2024; Pu, Yang, Li, & Guo, 2024). For instance, Pu, Yang, Li, and Guo (2024) used MLLMs to generate reports related to site conditions and safety hazards, demonstrating the potential for accelerating the construction inspection process.

Although MLLMs have been successfully applied in the construction field, the following research gaps remain. First, previous studies focused on single images and did not utilize multiple video frames, which limited their ability to recognize construction equipment tasks in a temporal context. Specifically, it remains unclear whether MLLMs, when provided with multiple video frames, will only offer information such as the color, location, and type of construction equipment as in traditional image captioning methods. Additionally, it is uncertain whether MLLMs can detect operations or significant events involving construction equipment within specific frames. Second, while MLLMs could recognize simple and representative tasks (e.g., the earthmoving task of an excavator), they had limitations in recognizing more complex tasks. This highlights a knowledge gap in how to enhance MLLM's ability to recognize complex tasks. Regarding the text prompt, it is important to determine what specific information should be incorporated into the prompts to accurately recognize distinct construction equipment tasks. Similarly, in visual prompt engineering, understanding how to process and emphasize visual inputs is essential for better directing the attention to construction equipment.

To address these challenges, the proposed framework employs a proprietary model capable of processing a long length of tokens, and an MCQ dataset was created to clarify the scope and depth of information that MLLMs can provide for construction equipment monitoring. Moreover, this study introduces two approaches aimed at improving the recognition of complex tasks: the idle-to-task approach for optimizing text prompts and the cropped visual inputs approach for refining visual inputs.

3 | METHODOLOGY

3.1 | Overall framework

The proposed framework aims to automatically analyze video data from construction sites to collect information about the location, idle state (i.e., idle or non-idle), and detailed operational tasks being performed by the

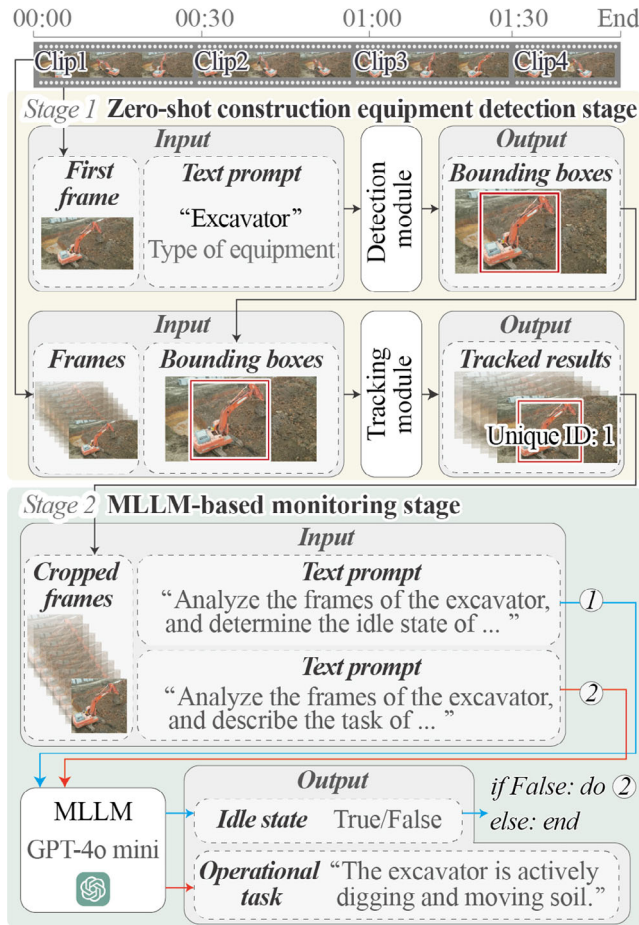


FIGURE 1 The overall framework consists of the zero-shot construction equipment detection stage and the MLLM-based monitoring stage.

construction equipment over time. To process the multiple video frames over a long duration efficiently and reduce memory usage, the proposed framework divides the video into 30-s clips, setting each clip as the input unit to recognize construction equipment operations. This duration was selected because typical cycle times of construction equipment operations, such as those of excavators, generally fall within a 30-s window, allowing meaningful operational tasks to be captured (Sağlam & Bettemir, 2018), while also helping reduce memory usage and enabling consistent batching of visual inputs for MLLM processing.

The framework consists of two stages (as shown in Figure 1): (i) the zero-shot construction equipment detection stage and (ii) the MLLM-based monitoring stage. The zero-shot construction equipment detection stage automatically detects and tracks the location of construction equipment within the clip (see Section 3.2). This stage includes two modules: a detection module and a tracking module. The detection module takes the first frame in the clip and the text prompt, the class of the construction equipment to be detected, as input. This detection module

then recognizes the objects (i.e., construction equipment) and generates their bounding boxes as output. The tracking module receives the bounding boxes and all frames of the clip as input. This tracking module then tracks the objects and generates masks and unique IDs as output. In the MLLM-based monitoring stage, the text prompt and the cropped frames from the tracked results are used as input to provide information about the construction equipment tasks. Notably, the idle states and operational tasks are recognized separately as the MLLM is run twice according to the idle-to-task approach (see Section 3.3).

3.2 | Zero-shot construction equipment detection stage

The zero-shot construction equipment detection stage automatically detects and tracks construction equipment within a clip. This stage provides a crucial foundation for refining data in the subsequent stage, where construction equipment tasks are analyzed in more detail. The zero-shot construction equipment detection stage consists of two modules: the detection module and the tracking module.

The detection module identifies the location and class of construction equipment by generating bounding boxes, which help track and analyze the construction equipment in the subsequent stage. The detection module utilizes Florence-2, a State-of-the-Art (SotA) object detection model developed by Microsoft, which can handle a variety of vision tasks and vision-language tasks, such as caption generation, object detection, visual grounding, and image segmentation (Xiao et al., 2024). Specifically, Florence-2 is capable of open vocabulary object detection, meaning it can detect objects described by a text prompt without being limited to predefined classes. Trained on approximately 126 million image-text pairs, it can detect domain-specific objects like construction equipment using ZSL. Florence-2 takes a single image and a text prompt as input and generates the corresponding bounding boxes related to a text prompt as output through its specialized architecture (see Figure 2a). The inputted image is the first frame in the clip, which is fed into the DaViT vision encoder and transformed into visual token embeddings. Visual token embeddings represent image features in vector form, making them interpretable by the transformer-based model. For the text prompt, the task description "< open vocabulary detection task >" is included, along with the input of construction equipment classes such as excavator, dump truck, and loader. Notably, the text prompt for construction equipment is designed to include word-level category labels instead of full sentences or detailed descriptions. This approach is consistent with the training data, as datasets such as Object365 (Shao et al., 2019), which are

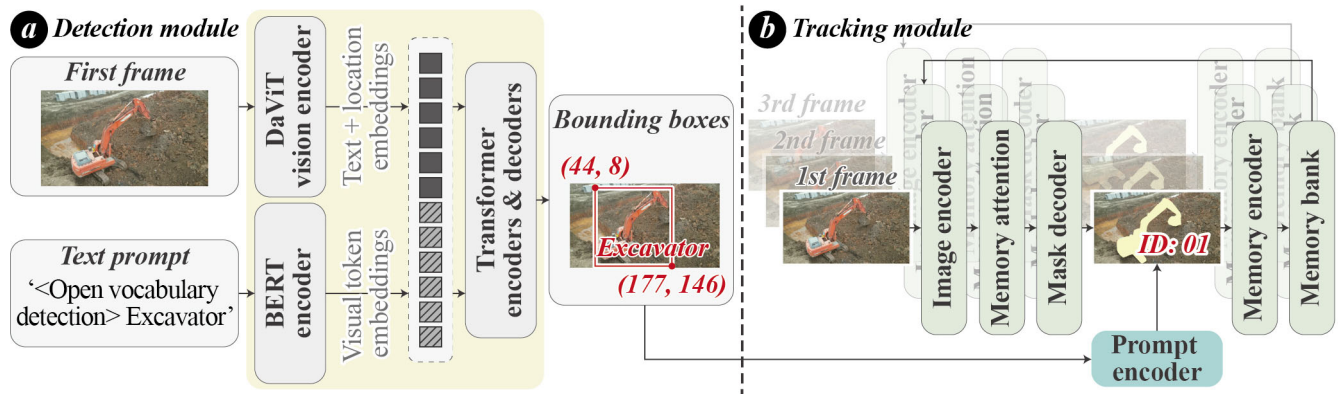


FIGURE 2 The zero-shot construction equipment detection stage consisting of a detection module (left) and a tracking module (right). The detection module utilizes the Florence-2 model, and the tracking module utilizes the Segment Anything Model 2 (SAM2).

integral to the dataset used to train Florence-2 (FLD-5B), primarily employ word-level annotations. The text prompt is then fed into BERT (i.e., encoder), which converts it into text and location embeddings. Location embeddings represent the coordinates of the box and are derived from location tokens added to the tokenizer vocabulary list (T. Chen et al., 2022). The visual token embeddings, along with the text and location embeddings, are processed by a transformer-based multi-modal encoder-decoder to generate the final output. The final output includes the coordinates of the bounding box, represented by the x and y values of the top-left and bottom-right corners, and the label of the detected construction equipment class. If no construction equipment is detected in the first frame, the clip is considered to contain no equipment, preventing unnecessary model operations and saving computing resources across the entire framework.

The tracking module tracks the detected construction equipment across the sequential frames of the clip, generating unique IDs and bounding boxes to continuously monitor the location of the equipment. For the tracking module, SAM2 is employed since it enables real-time, continuous tracking of target objects across multiple frames (Ravi et al., 2024). It demonstrates high performance in tracking multiple objects simultaneously within video sequences while maintaining a small size with 38.9 million parameters. It is particularly useful in dynamic and complex environments such as construction sites, as it can maintain tracking continuity even when the object is temporarily obscured or leaves and re-enters the scene. In addition, SAM2 has been trained on the SA-V dataset, which includes over 51,000 videos with mask annotations, giving it the potential to effectively track domain-specific objects such as construction equipment. To track construction equipment using SAM2, the bounding boxes of the equipment detected by Florence-2 in the first frame are input into SAM2 (see Figure 2b). SAM2 generates masks

for the construction equipment in the given frame based on the input bounding boxes, and these masks are stored in a memory bank. The memory bank efficiently stores and manages information from previous frames. In subsequent frames, SAM2 uses the memory attention module to reference the memory from previous frames and predict the masks for the current frame. During this process, the memory attention module selectively utilizes the most relevant previous information in relation to the current frame. The mask prediction results for each frame are stored back in the memory bank and are used for tracking in the following frames. This process is repeated throughout the entire video, ensuring continuous tracking of the construction equipment. As a result, SAM2 outputs a mask and unique ID for the construction equipment in each frame of a 30-s clip. Notably, the bounding boxes and unique IDs generated around the masks in the last frame are carried over to the subsequent clip, allowing for continuous recognition and tracking of the same object across different 30-s clips.

3.3 | MLLM-based monitoring stage

The MLLM-based monitoring stage identifies the idle states of construction equipment and automatically recognizes operational tasks based on the tracked results provided by the zero-shot construction equipment detection stage. For the MLLM-based monitoring stage, the proprietary model “GPT-4o mini” is selected due to its large context window, which can handle long token inputs, including both visual inputs (i.e., multiple cropped frames) and text prompts simultaneously. Moreover, it has high inference speed, low API cost, and can maintain reliable output quality compared to other MLLMs. The visual inputs and text prompts provided to the MLLM were configured based on the prompt guidelines from Google and OpenAI (Google AI for Developers, 2024; OpenAI, 2024).

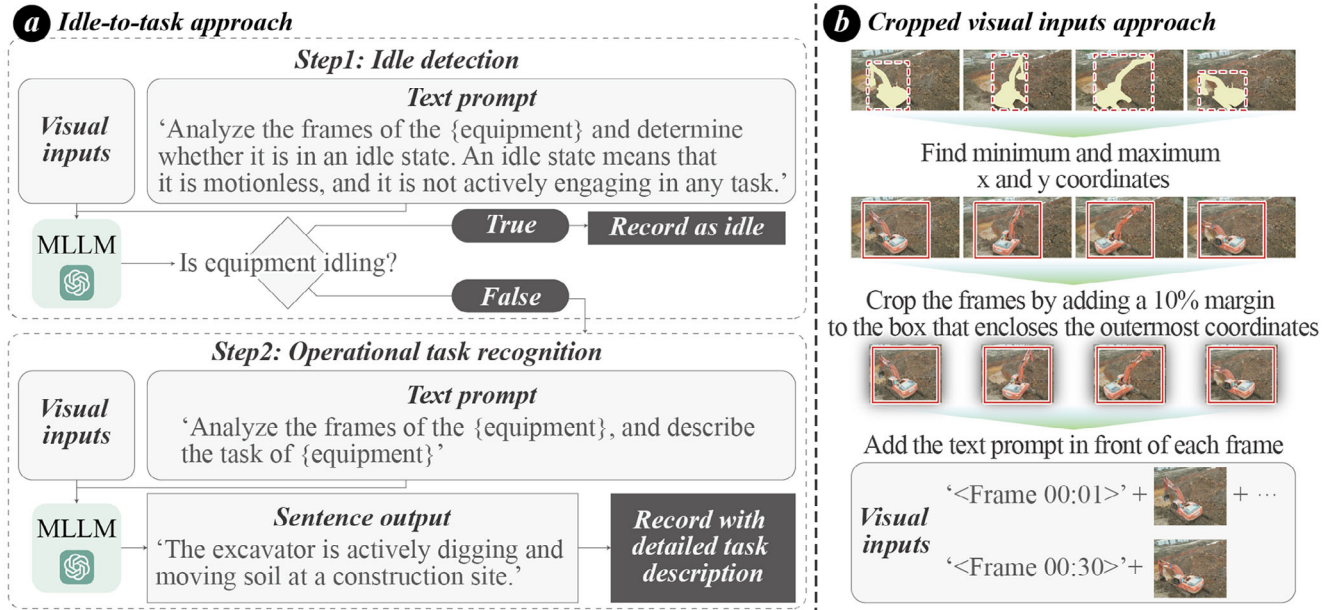


FIGURE 3 MLLM-based monitoring stage for recognizing construction equipment tasks with the idle-to-task approach (left) and the cropped visual inputs approach (right).

Specifically, the images were base64-encoded and resized so that their dimensions did not exceed 512×512 pixels. Additionally, the visual inputs were placed before the text prompt. To accurately recognize the construction equipment tasks, two approaches are proposed. For the text prompt, the idle-to-task approach is applied, and for the visual inputs, the cropped visual inputs approach is applied.

First, the idle-to-task approach is proposed to recognize construction equipment tasks in two steps (see Figure 3a). In the first step, frames are provided as visual inputs along with a text prompt that asks about the idle state. By focusing exclusively on the idle state in the first step, the idle-to-task approach establishes a foundation for analyzing non-idle equipment operations in the second step. Specifically, the text prompt includes a definition of the idle state and the class of construction equipment used in the zero-shot construction equipment detection stage. An example of this text prompt is: “Analyze the frames of the {equipment} and determine whether it is in an idle state. An idle state means that it is motionless, and it is not actively engaging in any task.” The MLLM returns a binary output regarding the idle state, and the response format in the API function is used to strictly limit the output to a Boolean type (True or False). If the idle state is not detected, the second step is executed. The MLLM uses the same visual inputs (i.e., frames) and text prompt as in the first step. The MLLM then generates a sentence output that provides a detailed description of the construction equipment task. The text prompt includes the class of construction equipment used in the zero-shot construction equipment

detection stage. An example of this text prompt is: “Analyze the frames of the {equipment} and describe the task of the {equipment}.” Similarly, the response format in the API function is used to ensure the output is in sentence format.

Second, the cropped visual inputs approach is a form of visual prompt engineering designed to help the MLLM recognize construction equipment tasks more precisely. In this approach, only the area related to the construction equipment is used as visual input, meaning that specific regions containing the equipment are cropped from all frames in the clip. Specifically, based on the tracked results provided by the zero-shot construction equipment detection stage, the region where the construction equipment appears is extracted from each frame by a unique ID (see Figure 3b). To do this, the minimum and maximum x and y coordinates of the masks in all frames of the 30-s clip are calculated. A bounding box is then defined around these outermost coordinates, with a 10% margin added in all directions to crop each frame in the clip. The purpose of this approach is to remove unnecessary background information and provide visual inputs that allow the MLLM to focus on the construction equipment tasks. Compared to non-cropped visual inputs of the same resolution, the cropped visual inputs contain more focused information about the construction equipment, enabling the MLLM to perform a more detailed analysis. Moreover, by cropping frames based on the outermost coordinates, this approach preserves contextual information more effectively compared to using only bounding boxes. For example, in a scenario where an excavator is loading soil onto a truck, if only the bounding boxes related



to the excavator are used as input, contextual information about the truck may be omitted. Additionally, a text prompt, such as “< Frame 00:01 >,” was added in front of each frame to indicate the time information, enabling the model to better understand the temporal context and accurately analyze the events across consecutive frames. Accordingly, the combined prompt consisting of visual inputs and the text instruction was structured as follows: “< Frame 00:01> {base64-encoded image} < Frame 00:02 > {base64-encoded image} ... Analyze the frames of the {equipment}....”

4 | EXPERIMENTS

4.1 | Comparative analysis of construction equipment detection

This section aims to evaluate whether the zero-shot construction equipment detection stage can effectively detect various types of construction equipment without domain-specific training data by utilizing ZSL. For this purpose, a comparative analysis was conducted between Florence-2 and YOLOv8. Additionally, the performance of YOLOv8 under a limited data scenario was analyzed. All experiments were conducted on a local machine with a 12th-generation Intel Core i7-12700KF CPU, NVIDIA GeForce RTX 3060 GPU, 32 GB RAM, using PyTorch 2.3.1 and CUDA 12.1.

4.1.1 | Dataset and experimental setup

Regarding the dataset, the Alberta Construction Image dataset (ACID) (Xiao & Kang, 2021; Xiao et al., 2022), which contains images from various construction sites, was used. The ACID dataset consists of 10,000 images and includes ten types of construction equipment (see Table 1).

Regarding the comparative models, the supervised learning model YOLOv8 was used. YOLOv8 was selected due to its ability to achieve both real-time detection performance and high accuracy. Additionally, to simulate a limited data scenario, various sizes of training sets were configured. Specifically, the model was trained with training sets consisting of 1000, 2000, 4000, and 8000 images. These training set sizes were chosen to reflect the practical challenges of data collection and labeling, starting with 1000 images (approximately 100 images per class) and gradually increasing the dataset size to analyze the trend of performance improvement. For the validation set, 2000 images were used, and both the training and validation sets were organized to ensure an equal distribution of the 10 classes. Florence-2 employs open

TABLE 1 Distribution of bounding boxes by class.

	Training set				Val. set
Number of samples	1000	2000	4000	8000	2000
Dump truck	222	463	937	2980	702
Excavator	216	445	902	2264	505
Wheel loader	152	288	620	1447	347
Backhoe loader	107	212	456	1128	264
Grader	111	227	453	1054	263
Compactor	127	239	494	966	234
Dozer	111	232	462	879	235
Mobile crane	120	236	463	870	237
Cement truck	103	211	430	650	201
Tower crane	104	213	301	389	77

vocabulary detection, which requires prior knowledge of the construction equipment present in the images. To address this requirement, the construction equipment classes from the ground truth labels were used as a text prompt. For example, if the ground truth labels included excavator and truck, the model was run twice, each time with “excavator” and “truck” entered as separate text prompts. It is worth noting that FSL was not included for comparison because it requires a training phase using extensive labeled data from related classes, which does not align with the goal of evaluating models under limited data conditions (J. Kim & Chi, 2021).

To systematically investigate the semantic boundaries of ZSL capabilities, an experiment was conducted by varying the descriptive detail in text prompts. Since ZSL capabilities rely on semantic understanding of input prompts, it is important to assess whether additional descriptive information, such as color or specific names, could affect detection accuracy. Specifically, the experiment explored four distinct text prompt configurations: (i) construction equipment class for baseline (e.g., excavator), (ii) dominant color combined with the equipment class (e.g., yellow excavator), (iii) manufacturer’s model name (e.g., Caterpillar 336D), and (iv) a combination of dominant color and manufacturer’s model name (e.g., yellow Caterpillar 336D). Toward this end, dominant colors and manufacturer’s model names were manually assigned to each equipment instance in the validation set. Additionally, Florence-2 was run separately for each text prompt configuration. The detected bounding boxes from each individual execution were then assigned class IDs, followed by the Non-Maximum Suppression (NMS) to eliminate redundant and overlapping detections.

Regarding the evaluation metrics, precision, recall, and F1-score, which are commonly used to assess the performance of object detection models, were used. Precision



TABLE 2 Comparison of precision, recall, and F1-score for You Only Look Once (YOLO) v8 and Florence-2 models.

Model	Train samples	Precision (%)	Recall (%)	F1-score (%)
YOLOv8n	1000	67.2	56.4	61.3
	2000	82.1	70.6	75.9
	4000	87.3	79.1	83.0
	8000	93.0	83.0	87.7
YOLOv8s	1000	72.6	59.8	65.6
	2000	84.4	74.2	79.0
	4000	90.0	81.5	85.5
	8000	90.3	87.7	89.0
YOLOv8m	1000	77.2	64.4	70.2
	2000	84.4	76.8	80.4
	4000	89.7	84.3	86.9
	8000	93.0	90.1	91.6
Florence-2 base	0	83.6	78.5	81.0
Florence-2 large	0	87.1	77.8	82.2

represents the proportion of correctly identified objects out of all detected objects, while recall represents the proportion of actual objects that were successfully detected. The F1-score, which is the harmonic mean of precision and recall, provides a balanced evaluation of the two metrics. For YOLOv8, the Intersection over Union (IoU) threshold was set at 0.45, and any detection with a confidence score higher than 0.25 was considered a true positive. In contrast, Florence-2 does not provide confidence scores like supervised learning models (e.g., YOLOv8). Therefore, only the IoU threshold was used to evaluate Florence-2's performance.

4.1.2 | Zero-shot detection performance analysis

Table 2 shows the precision, recall, and F1-score of Florence-2 and YOLOv8 based on the size of the training samples. The values are derived by averaging the precision, recall, and F1-score across all classes. YOLOv8n, YOLOv8s, and YOLOv8m have 3.2, 11.2, and 25.9 M parameters, respectively, while Florence-2 base and Florence-2 large contain 0.23 and 0.77 B parameters. The experimental results indicate that Florence-2 base and Florence-2 large achieved an F1-score of 81.0% and 82.2%, respectively. On the other hand, YOLOv8n presented the lowest F1-score of 61.3% when trained with 1000 samples, while YOLOv8m achieved the highest F1-score of 91.6% when trained with 8000 samples. The YOLOv8 models showed an improvement in precision, recall, and F1-score as the size of the model and the training samples increased,

TABLE 3 Performance of the Florence-2 large across different text prompt configurations.

Text prompt configuration	Precision (%)	Recall (%)	F1-score (%)
Baseline	87.1	77.8	82.2
Color	87.8	79.2	83.3
Manufacturer's model	57.1	53.8	55.4
Color + Manufacturer's model	76.2	71.3	73.7

which is consistent with the neural scaling law (Hestness et al., 2017). These results suggest that supervised learning models can deliver high performance when provided with sufficient data. In contrast, Florence-2 large outperformed the YOLOv8 models when the training samples were fewer than 2000. This demonstrates that Florence-2, which utilizes ZSL, can offer practical advantages in limited data scenarios.

Table 3 shows the performance of Florence-2 large across different text prompt configurations. The baseline prompt, using only construction equipment class as text prompt, achieved an F1-score of 82.2%, serving as a reference point for semantic flexibility analysis. When color information was included in the text prompt (e.g., yellow excavator), the F1-score increased to 83.3%. However, introducing manufacturer's model names reduced performance, achieving the lowest F1-score at 55.4%. The combined prompt of color and manufacturer's model name partially mitigated this reduction, reaching an F1-score of 73.7%. These results indicate that Florence-2 can effectively leverage simple semantic attributes, such as color, without substantial loss in performance. Conversely, the inclusion of highly specific, domain-oriented details like manufacturer's model names negatively impacts performance, suggesting practical semantic boundaries of Florence-2's ZSL capabilities within construction equipment contexts.

In conclusion, the results demonstrate that the performance of ZSL and supervised learning models varies depending on model size and the size of the training samples. The supervised learning models, such as YOLOv8, showed superior performance when there was sufficient training data and computational resources. In contrast, Florence-2, which utilizes ZSL, was particularly effective in limited data scenarios. Moreover, the results presented that overly specific prompts (e.g., manufacturer's model names) negatively impact its accuracy, while simple semantic attributes, such as color, can be effectively leveraged. These findings imply that while Florence-2 is highly suitable for immediate deployment in construction environments, practitioners should carefully consider



semantic specificity in prompt design to maximize its practical effectiveness.

4.2 | Performance evaluation of MLLM-based monitoring

This section aims to evaluate the performance of the MLLM-based monitoring stage. For this purpose, the MCQ dataset was created using YouTube videos featuring construction equipment, and the evaluation was conducted using the proprietary models GPT-4o mini and Gemini 1.5 Flash APIs. The MCQ dataset was designed to clarify the scope and depth of information that MLLMs can provide for construction equipment monitoring. Moreover, the sensitivity of the models to variations in idle state definitions was investigated by examining six variations in the text prompt designs.

4.2.1 | MCQ dataset design for MLLM-based monitoring

MCQ dataset was created to evaluate the MLLM-based monitoring stage. Since MLLMs that utilize ZSL can handle a wide range of dialogue-based tasks, evaluating them can be challenging. Accordingly, this study employed an MCQ dataset, which allows for simple and quick evaluations and is widely used in the computer vision field (Hendrycks et al., 2021; Wu et al., 2024). To create the MCQ dataset, a total of 52 videos were collected from YouTube. From each video, clips containing different scenes were extracted at intervals of less than 30 s, resulting in a total of 216 clips. The average length of each clip was approximately 28.5 s, with lengths ranging from 7 to 30 s. For each clip, questions with four possible answers were created, leading to a total of 1207 questions. Specifically, 782 questions (65.6%) and 213 questions (17.6%) focused on activities and equipment, respectively, while the remaining 202 questions (16.7%) addressed aspects such as counts, colors, and environment (see Figure 4). Moreover, the activity-related questions covered a diverse range of tasks, such as digging, loading, unloading, lifting, carrying, pushing, leveling, breaking, demolition, and idling, enabling the evaluation of performance across various construction equipment tasks. As a result, an average of 5.59 questions were generated per clip, with a minimum of three and a maximum of nine per clip.

The MCQ dataset consists of three types (see Table 4). While a prior study has focused on spatial inference questions—information that can be obtained from a single image—this study extends the evaluation framework by incorporating task inference and temporal event questions

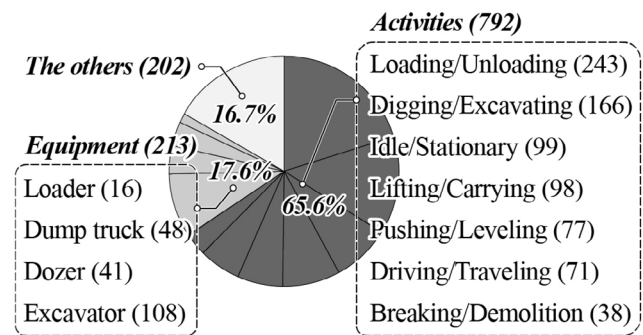


FIGURE 4 Breakdown of the Multiple Choice Question (MCQ) dataset by correct answer type.

(H. Chen et al., 2024). Specifically, the MCQ dataset was designed for understanding where equipment is located in relation to its surroundings (spatial inference), what actions and tasks the equipment is performing (task inference), and when critical events occur within the operational timeline (temporal event questions). Although these question types have limitations in evaluating comprehensive monitoring aspects related to human–equipment interactions or safety, they provide information regarding equipment productivity and operational states.

First, spatial inference questions describe the position, arrangement, and spatial relationships between construction equipment and other objects (e.g., soils, workers, site, and other equipment). These questions can be answered based on a single image. For example, questions may be asked about the positional relationship between an excavator and a dump truck, which construction equipment is visible in the scene, or the spatial positioning between a bulldozer and an excavator.

Second, task inference questions require reasoning about the activity, task, or purpose of construction equipment based on all the input frames. For example, questions might ask what task the excavator is performing (e.g., earthmoving, transporting material) or what the likely purpose of the bulldozer in the background is.

Third, temporal event questions require understanding the timing, sequence, or specific frames where a particular event or action occurs. For instance, questions may be asked when the excavator begins loading soil into the dump truck, at which frames the excavator is lifting a rock from the ground, when the loader appears to be idle, or which frame shows the first appearance of the dump truck.

To compare the performance of MLLMs and determine the most appropriate model, three cases were compared: (i) Gemini 1.5 Flash (video), (ii) Gemini 1.5 Flash (frame), and (iii) GPT-4o mini. Gemini 1.5 Flash was selected as a comparative model because, among various proprietary models, it has the most similar API cost and model size to GPT-4o mini. In particular, GPT-4o mini costs \$0.15 per

TABLE 4 Examples of three types of questions in the Multiple Choice Question (MCQ) dataset.

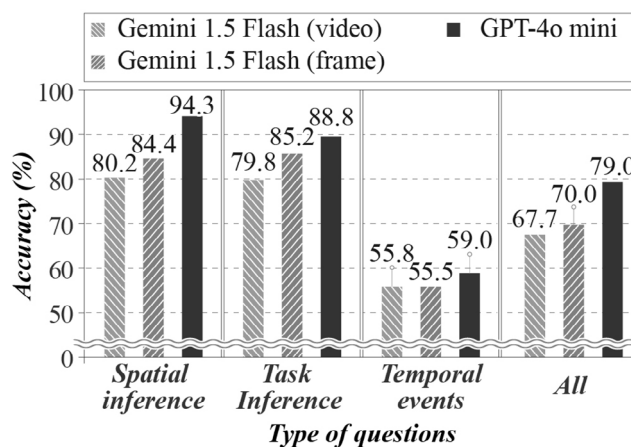
Type	Examples
Spatial inference (263 questions)	Q. What type of equipment is primarily shown in the images? (A) Bulldozer (B) Excavator (C) Crane (D) Loader
	Q. What is the position of the excavator in relation to the dump truck during the loading process? (A) Beside the dump truck (B) Behind the dump truck (C) In front of the dump truck (D) Above the dump truck
Task inference (331 questions)	Q. What is the excavator doing in the sequence of images? (A) Moving soil (B) Breaking rocks (C) Lifting heavy materials (D) Transporting rocks
	Q. What activity does the bulldozer perform after the truck is loaded? (A) Starts digging (B) Pushes material (C) Moves to another location (D) Loads more material
Temporal event (613 questions)	Q. What activity is NOT observed in the Frame 00:18–00:20? (A) Excavator in the middle is drilling (B) Excavator on the left is digging (C) Multiple trucks in the background are stationary (D) Dump truck is being loaded with the material from the excavator
	Q. In which frame does the excavator idle with no movement? (A) Frame 00:01–00:10 (B) Frame 00:11–00:20 (C) Frame 00:21–00:30 (D) Excavator is actively working in all frames

1 million input tokens, while Gemini 1.5 Flash costs \$0.10 per 1 million input tokens, making both models among the most cost-effective options (Artificial Analysis, 2024) and enabling improved scalability in real-world applications. Since Gemini 1.5 Flash can take both video files and multiple frames as visual inputs, performance was compared based on the input format. In the case of Gemini 1.5 Flash (video), the video clip file was extracted at 1 frames per second (fps), resulting in a similar API cost as using multiple frames as visual inputs. As a result, Gemini 1.5 Flash (video) used a single video clip file as the visual input, while Gemini 1.5 Flash (frame) and GPT-4o mini used multiple frames extracted at 1-s intervals from the clips as their visual inputs. Notably, to evaluate only the MLLM-based monitoring stage, the original non-cropped frames were used. Meanwhile, to eliminate the inherent randomness of the MLLM, the temperature hyperparameter was set to 0, which controls the randomness in the model's output; lower values result in more deterministic and consistent responses.

4.2.2 | Evaluation of MLLM-based monitoring performance

Figure 5 shows the accuracy of the MLLMs on the MCQ dataset. The performance was evaluated based on three types of questions: spatial inference, task inference, and temporal event questions.

First, for the spatial inference questions, GPT-4o mini achieved the highest accuracy of 94.3%. On the other hand, Gemini 1.5 Flash (video) and Gemini 1.5 Flash (frame) recorded accuracies of 80.2% and 84.4%, respectively, showing that the model performed better when frames were

**FIGURE 5** Comparison of the accuracy of the MLLMs based on three types of questions: spatial inference, task inference, and temporal event questions.

used as visual inputs rather than video. All cases achieved over 80% accuracy, indicating that MLLMs can reliably provide information about the location and appearance of construction equipment.

Second, for the task inference questions, a similar pattern to the spatial inference results was observed. GPT-4o mini achieved the highest accuracy at 88.8%. Notably, GPT-4o mini's accuracy dropped by 5.5% for task inference questions, compared to spatial inference. This suggests that task inference, which requires reasoning across multiple frames, poses a greater challenge, compared to spatial inference, which can be inferred from a single frame. Overall, the models achieved between 79% and 89% accuracy, indicating that MLLMs can effectively capture information about construction equipment tasks from multiple video frames.

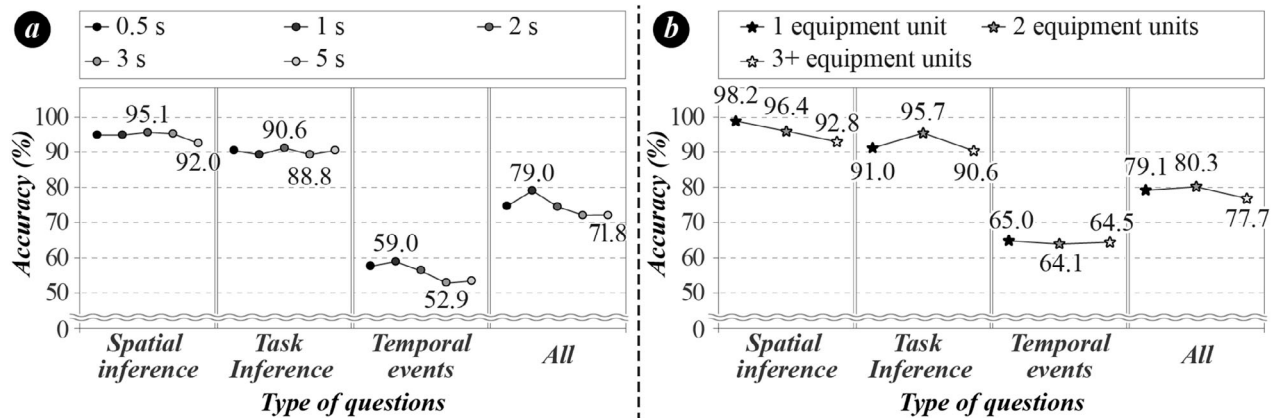


FIGURE 6 Comparison of the accuracy of the MLLM (GPT-4o mini) with different visual input intervals (left), and the impact of site complexity, measured by the number of construction equipment in the clip (right).

Third, for the temporal event questions, there was a noticeable drop in accuracy. GPT-4o mini recorded an accuracy of 59.0%, which was 35.3% lower than for spatial inference questions and 29.8% lower than for task inference questions. Gemini 1.5 Flash (video) and Gemini 1.5 Flash (frame) recorded 55.8% and 55.5%, respectively, making temporal event questions the category with the lowest accuracy. These results suggest that MLLMs have limitations in recognizing specific events through frame-to-frame comparisons. Specifically, the temporal variations in construction equipment videos captured by stationary cameras are often subtle, leading to minimal differences in the visual embeddings between frames, which in turn limits the recognition of events in specific frames. With accuracy below 60% across all cases, these findings highlight the challenges MLLMs face in collecting information based on specific frames. This indicates limitations in achieving precise temporal accuracy at the second-by-second level, particularly for predicting the earthmoving cycle, which involves tasks such as digging and loading.

When considering all question types, GPT-4o mini outperformed the other models with an overall accuracy of 79.0%, followed by Gemini 1.5 Flash (video) at 67.7% and Gemini 1.5 Flash (frame) at 70.0%. These results indicate that performance can vary across proprietary models for construction equipment monitoring. However, due to differences in API costs and data policies, model selection cannot be based on performance alone. The results suggest that site managers must consider multiple factors comprehensively, not just performance, when selecting an MLLM for construction equipment monitoring.

Figure 6a shows the accuracy of GPT-4o mini across different visual input intervals. The results showed that the model achieved an accuracy exceeding 92% for spatial inference questions, which indicates that extensive temporal context from multiple frames is not essential for this

task. For task inference questions, the performance varied by a maximum of 3.3% across the different intervals. In contrast, the accuracy for temporal event questions remained low, with a maximum of 59%. Even when a 0.5-s interval was used, the accuracy did not improve compared to when a 1-s interval was used, implying that simply increasing the number of frames does not necessarily enhance performance in capturing specific temporal events.

Figure 6b shows the accuracy of GPT-4o mini according to the number of construction equipment units present in each clip, which reflects varying levels of environmental complexity. The model achieved the highest accuracy when only one piece of equipment was visible, with 98.2% for spatial inference, 95.7% for task inference, and 65.0% for temporal event questions. When two equipment units were present, the accuracy slightly decreased to 96.4% for spatial inference, 91.0% for task inference, and 64.5% for temporal events. In scenes with three or more equipment units, the performance further declined to 92.8%, 90.6%, and 64.1% for each question type. Overall, the accuracy dropped from 79.1% with one equipment unit to 77.7% with three or more, suggesting that increased visual complexity may challenge the model's reasoning capability. Nevertheless, the model consistently maintained high accuracy for spatial and task inference questions, even in more complex scenes, indicating its potential applicability in varied construction site environments.

In conclusion, the results demonstrate the significant potential of MLLM-based monitoring for construction equipment, achieving up to 79.0% accuracy on the MCQ dataset. The models performed particularly well on spatial and task inference questions, suggesting their value in recognizing the location and operational tasks of construction equipment. However, accuracy was lower for temporal event questions, indicating the need for further improvements, such as fine-tuning (Y. Jung et al., 2024), to achieve second-by-second-level analysis. Meanwhile, the newly

TABLE 5 Six different prompt designs defining idle states for the idle-to-task approach, and the corresponding accuracies.

Case	Text prompt	Acc. (%)
P_0	Analyze the frames of the {equipment} and determine whether it is in an idle state. (baseline)	88.64
P_S	... An idle state means that it is stationary	90.91
P_M	... An idle state means that it is motionless	92.42
P_{NA}	... An idle state means that it is not actively engaging in any task	89.39
P_{S+NA}	... An idle state means that it is stationary, and it is not actively engaging in any task	91.67
P_{M+NA}	... An idle state means that it is motionless, and it is not actively engaging in any task	93.18

developed MCQ dataset allowed for a systematic evaluation of the performance of the MLLM in construction equipment monitoring, which is expected to contribute to future research in this area.

4.2.3 | Sensitivity to the definition of idle state

Table 5 presents a comparative analysis of idle state detection accuracy across six different definitions of “idle” used in the text prompt of the idle-to-task approach. Previous studies identified the idle state of construction equipment based on frame-difference thresholds applied to the pixels within the bounding box of detected equipment (C. Chen et al., 2023; Zou & Kim, 2007). Guided by this principle, six distinct text prompts were formulated to examine their influence on the accuracy of GPT-4o mini in classifying idle states (see Table 5). The experiment utilized clips from two sources: (i) 96 clips were created by cropping ACID dataset images to each equipment’s bounding box and duplicating each image into 30 identical frames to simulate a 30-s idle scenario; (ii) 36 clips representing non-idle scenarios were selected from the MCQ dataset (see Section 4.2.1) and similarly cropped to the equipment’s bounding box, consistent with the idle-to-task approach (see Section 3.3).

The results indicate that the baseline prompt definition (P_0) yielded the lowest accuracy at 88.64%. Adding either spatial cues, such as “stationary” (P_S , 90.91%) and “motionless” (P_M , 92.42%), or functional cues, such as “not actively engaging in any task” (P_{NA} , 89.39%), improved the accuracy. Furthermore, combinations of spatial and functional definitions resulted in even higher accuracy, with P_{S+NA} achieving 91.67% and P_{M+NA} attaining the highest accuracy of 93.18%. These findings demonstrate that combining the concepts of being physically motionless and functionally inactive provides the most precise description for identify-

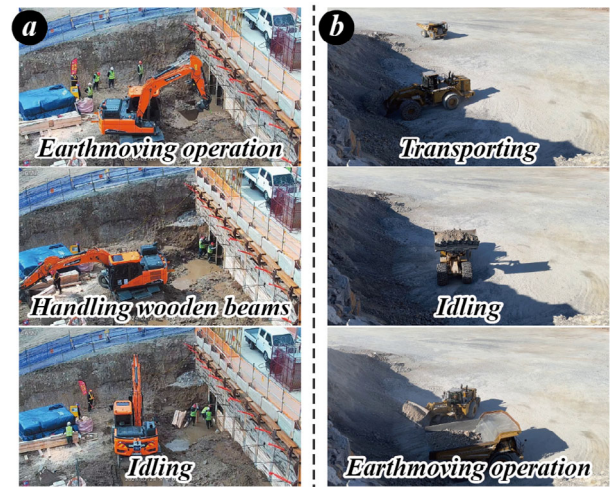


FIGURE 7 Real-world case study videos: (a) excavator operations at a Busan construction site and (b) loader and dump truck coordination from YouTube.

ing idle states. In other words, the superior performance of the P_{M+NA} suggests that the term “motionless” provides a stricter, less ambiguous signal of zero movement than “stationary,” thereby allowing the model to more decisively separate idle from non-idle frames.

4.3 | Real-world case study

In this section, the proposed framework was applied to a video captured at an actual construction site to verify its effectiveness. Specifically, an ablation study was designed to verify the effectiveness of two approaches introduced to improve the performance of MLLM-based monitoring: the idle-to-task approach and the cropped visual inputs approach. Furthermore, the framework’s performance was examined using videos featuring multiple types of construction equipment.

4.3.1 | Implementation details

Figure 7 shows two videos for the case study. Figure 7a shows the first video that was captured by CCTV at a construction site in Busan. The video has a framerate of 30 fps, a resolution of 1920×1080 , and a length of 13 min and 25 s. It includes earthmoving operations, such as excavating and transporting soil from a filled pit and handling wooden beams by loading them into the bucket and delivering them to workers. Additionally, the last 3 min of the video show the excavator in an idle state, making it ideal for assessing whether the proposed framework can effectively analyze changes in construction equipment tasks



over time. The video was divided into 30-s intervals, producing a total of 27 clips. Figure 7b is the second video sourced from YouTube recorded at 30 fps, with a resolution of 1920×1080 , and a duration of 4 min and 30 s. It captures the earthmoving operations of a loader, including the loading of soil into a dump truck and the subsequent transportation of the truck, thereby illustrating the interaction among multiple construction equipment. This video was also divided into 30-s intervals, yielding a total of nine clips.

Regarding the models used for the proposed framework, they were selected based on the results of prior experiments. For the detection module, Florence-2 large, which showed the highest performance using ZSL, was utilized. For open vocabulary object detection, the text prompt “excavator” was used for the first video, while “wheel loader” and “dump truck” were used for the second video, resulting in the detection of a bounding box for the first frame of each clip. The tracking module employed SAM2, which recognized the region containing the excavator in all frames of the clip and extracted that region into cropped frames. In the MLLM-based monitoring stage, GPT-4o mini, which achieved the highest accuracy on the MCQ dataset, was used, and the cropped frames based on the tracked results were used as the visual inputs.

Regarding the evaluation process, the binary output for the idle state and the sentence output for the operational task, both generated by the MLLM-based monitoring stage, were analyzed to calculate the accuracy for each clip. First, evaluating the idle state was straightforward, as it was defined by the absence of movement. For instance, if no pixel changes were observed in the tracked results of the construction equipment across all frames, the clip was labeled as idle. Second, the description of the operational task was manually evaluated. Due to the complexity of interpreting sentence-level semantics and context, human-based evaluation provided greater reliability. Any inaccuracies in the content of the description were marked as incorrect predictions. For example, if a 30-s clip showed an excavator scooping dirt and loading it back onto the ground, but the framework described it as “*The excavator scooping soil from the ground and loading it onto a truck,*” it would be considered an incorrect prediction.

4.3.2 | Ablation study: Assessing idle-to-task and cropped inputs approaches

An ablation study was conducted to verify the effectiveness of the idle-to-task approach and the cropped visual inputs approach. The goal was to analyze the impact of each method on the performance of the MLLM-based monitor-

ing stage and to assess their individual contributions. To this end, the following three scenarios were additionally evaluated for the first video: (A) when neither the idle-to-task approach nor the cropped visual inputs approach was applied, (B) when the cropped visual inputs approach was not applied, and (C) when the idle-to-task approach was not applied.

- A. Scenario without both idle-to-task approach and cropped visual inputs approach: This served as a baseline for comparing the contributions of the two approaches. In this case, the MLLM-based monitoring stage recognized the operational task immediately without detecting the idle state and used the non-cropped frames as visual inputs.
- B. Scenario without cropped visual inputs approach: In this case, the idle-to-task approach was applied, but non-cropped frames were used as visual inputs instead of cropped frames. Since the MLLM had to recognize tasks with the surrounding background included, this scenario allowed for evaluating the effectiveness of the cropped visual inputs approach, using only the regions extracted by the zero-shot construction equipment detection stage as visual inputs.
- C. Scenario without idle-to-task approach: In this case, the operational task was recognized immediately without detecting the idle state, allowing for an assessment of the effectiveness of the idle-to-task approach that distinguishes between idle and operational states.

Figure 8 shows a comparative analysis of recognized operational tasks across the different monitoring approaches. The timeline includes earthmoving operations, handling wooden beams, and idling, with the ground truth displayed at the top. First, in scenarios (A) and (B), where the cropped visual inputs approach was not applied, 16 out of the 27 clips were recorded as incorrect predictions. Specifically, between Clips 7 and 9, the task of handling wooden beams was incorrectly recognized as earthmoving operations. Additionally, the idle state starting from Clip 22 was misinterpreted as earthmoving operations. These results remained unchanged even when the idle-to-task approach was applied without the cropped visual inputs approach. In other words, when the zero-shot construction equipment detection stage was bypassed, incorrect predictions continued to occur, and the idle-to-task approach alone was not sufficient to significantly improve accuracy. Second, in scenario (C), where the cropped visual inputs approach was applied but the idle-to-task approach was not, 11 out of 27 clips were recorded as incorrect predictions, showing improved performance. Specifically, the task of handling wooden beams was correctly recognized between Clips 7 and 9 and between Clips 19 and 20. How-

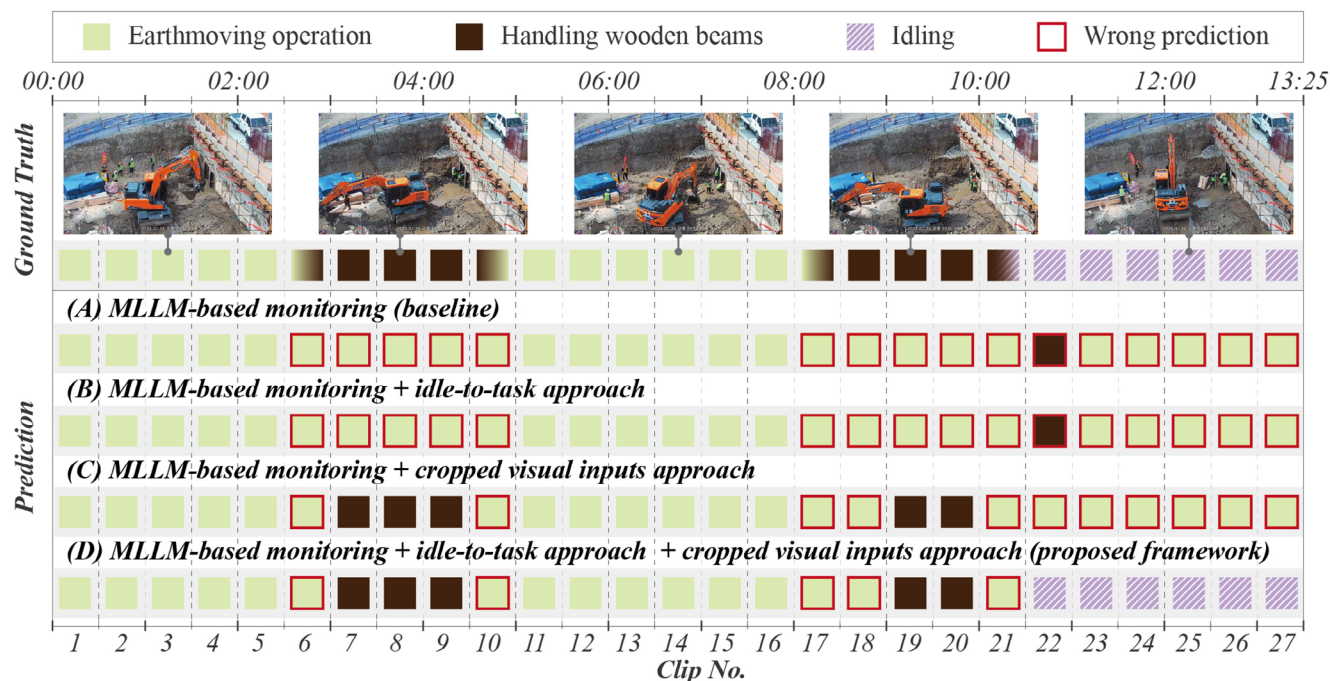


FIGURE 8 Results of the ablation study. The gradient in the ground truth labels represents the transition between tasks.

TABLE 6 Detailed responses of the proposed framework in the case study.

Clip No.	Idle state	Operational task
Clip 5	False	The excavator is tasked with digging and moving soil at a construction site, effectively scooping up dirt and positioning itself to clear the area for foundation work
Clip 6	False	The excavator is engaged in lifting and moving materials, likely soil or debris, at a construction site, while workers coordinate the operation. (Incorrect prediction)
Clip 8	False	The excavator is tasked with lifting and moving wooden materials at a construction site, working in coordination with the workers
Clip 18	False	The excavator is actively engaged in digging and moving materials on a construction site, lifting and positioning its bucket to scoop up dirt or debris while workers assist in the operation. (Incorrect prediction)
Clip 27	True	–

ever, the idle state was not accurately recognized due to the use of a text prompt that did not include a detailed description of idling for the MLLM. These results suggest that using an appropriate prompt is crucial for accurately recognizing construction equipment tasks. Last, in the proposed framework, scenario (D), where both the idle-to-task approach and the cropped visual inputs approach were applied, only five out of 27 clips were marked as incorrect predictions, showing the highest performance. Specifically, the idle state of the excavator depicted from Clip 22 was accurately identified. These findings indicate that the proposed framework is most effective where both approaches are used together.

Table 6 provides a detailed view of the idle state and operational tasks output by the proposed framework. These results suggest that the framework can recognize

various tasks, such as earthmoving operations like digging and moving soil, as well as lifting and moving wooden beams. Additionally, the sentence output, which includes elements such as soil, wooden materials, and workers, indicates that the proposed framework can analyze the context related to construction equipment. However, some tasks were inaccurately recognized in several clips. For example, in Clip 6, after the earthmoving operations ended, the rotation of the excavator's arm and body was not detected. This highlights a limitation in the MLLM-based monitoring stage's capability to focus on temporal events that occur in specific frames. Similar limitations were observed in the results for temporal event questions (see Section 4.2.2), suggesting that the framework has difficulty accurately processing tasks that occur only in specific frames within a sequence. Nonetheless, the pro-



posed framework successfully recognized the idle states and operational tasks (e.g., earthmoving operations, lifting and moving wooden beams), indicating that it can be effectively applied to automatically generate construction task reports or conduct productivity analyses based on the idle state, particularly in cases where minute-by-minute precision is required.

In conclusion, the case study showed that the proposed framework can effectively monitor various construction equipment tasks on a construction site. In particular, the introduction of two approaches—idle-to-task approach and cropped visual inputs approach—contributed to accurately recognizing complex construction equipment tasks. The practical implications of the results suggest that this framework could contribute to the automation of construction equipment monitoring and the digitalization of site management. By installing cameras on a construction site and applying the proposed framework, information on construction equipment tasks can be automatically collected.

4.3.3 | Assessing framework on multiple construction equipment

This section evaluates the monitoring capability of the framework when multiple types of construction equipment operate simultaneously within the same video frame. Specifically, the performance of the proposed framework was evaluated using the second video featuring a wheel loader and a dump truck involved in earthmoving operations. Additionally, this analysis examines the impact of variations in the text prompt design on the accuracy of the MLLM-based monitoring stage, as well as compares Florence-2 large with a YOLOv8m trained on 8000 samples (see Table 2) to highlight the strengths and limitations of the ZSL approach. Toward this end, four distinct configurations were established for the evaluation (see Figure 9). In configuration (A), YOLOv8 was used for detection, and the idle-to-task approach was applied without explicitly defining what constitutes an idle state (i.e., baseline prompt definition (P_0), see Section 4.2.3). In configuration (B), YOLOv8 was also used for detection, but the idle-to-task approach was applied with an explicit definition of the idle state in the prompt (P_{M+NA}). In configuration (C), Florence-2 large was used for detection, and the idle-to-task approach was applied without explicitly defining the idle state (P_0). In configuration (D), Florence-2 large was used for detection, and the idle-to-task approach was applied with an explicit definition of idle in the prompt (P_{M+NA}).

Figure 9 shows a comparative analysis across the different configurations. The results showed the consistent

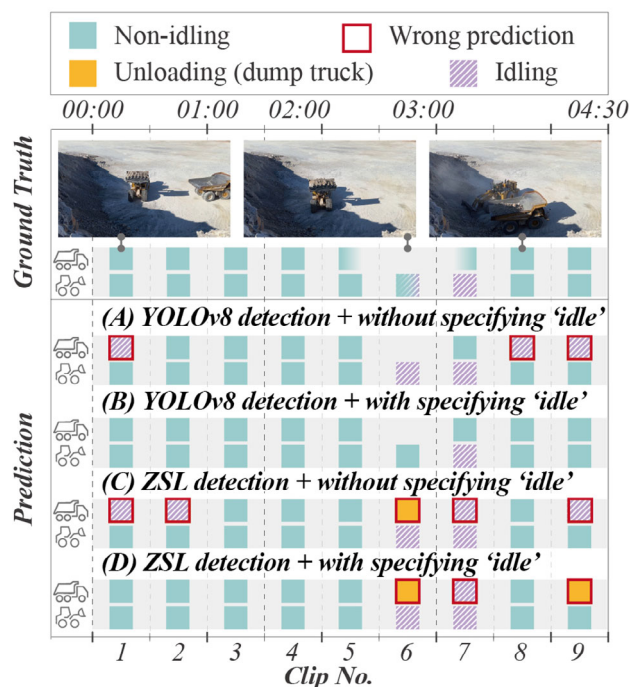


FIGURE 9 The results of the proposed framework with four distinct configurations. In Clip 6, it indicates that the dump truck has disappeared, and the gradient in the ground truth labels represents the transition between tasks.

recognition of the wheel loader's tasks across all four configurations. During Clips 6 and 7, the wheel loader exhibited an idle state by maintaining a stationary position while raising its bucket in preparation for loading, suggesting that the MLLM-based monitoring stage effectively distinguishes subtle operational differences in active and stationary contexts. Conversely, the performance varied for the dump truck across different configurations. When the idle state was not explicitly defined in the text prompt (i.e., configurations (A) and (C)), the MLLM-based monitoring stage misclassified the dump truck's task as idle states in the first two clips. Additionally, stationary moments when the dump truck was actively receiving soil (i.e., Clips 8 and 9) were incorrectly identified as idle. Conversely, explicitly defining the term "idle" in the text prompt (i.e., configurations (B) and (D)) improved idle recognition accuracy by clarifying the distinction between idle and actively engaged stationary states. Regarding the detection module performance, YOLOv8 consistently and robustly detected both equipment types, even when the dump truck partially moved out of frame (i.e., Clip 6). In contrast, Florence-2 large experienced occasional misclassification issues. For instance, in Clips 6 and 9, it incorrectly identified the dump truck as a wheel loader, which led the MLLM-based monitoring stage to incorrectly infer an "unloading" task that was not present. Nonetheless, Florence-2 large consistently and accurately detected

and tracked the wheel loader, resulting in reliable task recognition for this equipment throughout all video clips.

In conclusion, explicitly defining the idle state in the text prompt enhanced the accuracy of operational task recognition, particularly when distinguishing subtle differences between active engagement and idleness. The supervised learning models (i.e., YOLOv8) provided consistently robust performance, effectively handling multiple equipment types and partial occlusions. In contrast, the ZSL-based detection module exhibited accumulation errors due to sequential processing stages, whereby initial detection inaccuracies negatively influenced subsequent task recognition. Nonetheless, the proposed framework mitigates the cold-start issue by reducing reliance on extensive training data, thereby enabling rapid deployment in construction environments with limited data availability. Moreover, the accurate recognition of the wheel loader's operational tasks suggests that the framework reliably detects continuously operating equipment, underscoring its suitability for scenarios involving constant equipment use.

5 | DISCUSSIONS

5.1 | Practical implications

One of the key advantages of the proposed framework is its ability to function effectively without the need for training data by utilizing ZSL and MLLMs. Traditional supervised learning models require large volumes of labeled data to perform accurately, but in construction environments, collecting and labeling data are time-consuming, costly, and often impractical. This framework eliminates the reliance on labeled data, allowing it to be applied immediately to new sites. In particular, the framework is highly flexible, as it can recognize new construction equipment or tasks simply by changing the text prompts, without the need for training the model. Additionally, the framework can handle rare construction equipment tasks (e.g., handling wooden beams) by leveraging MLLM. These features enable easy adoption by small construction companies or projects with limited resources, as site managers can effectively monitor construction equipment tasks.

In the case study, the idle states and main operational tasks in the first video were recognized with 81.5% accuracy, indicating that the proposed framework could help automate the generation of construction equipment reports. Currently, creating daily work logs with manual observation is time-consuming and prone to human error. Using the proposed framework, task descriptions generated for each 30-s clip could be summarized by leveraging LLMs, providing a comprehensive understanding of equip-

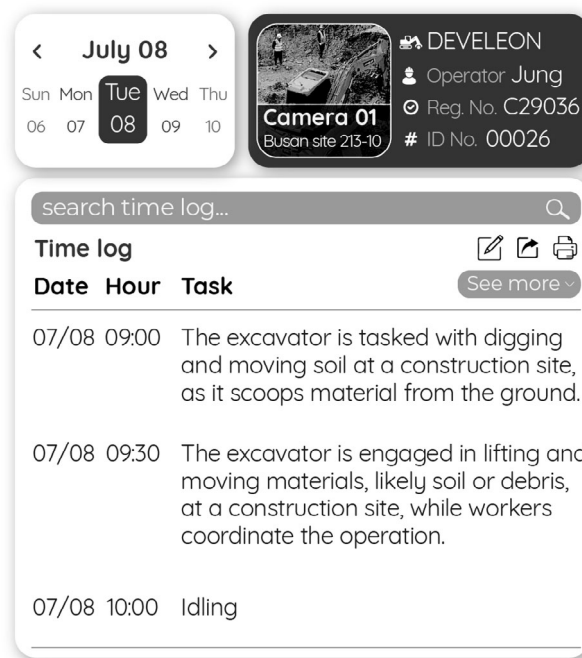


FIGURE 10 Example of the user interface for automating the report generation of the construction equipment task by using the proposed framework.

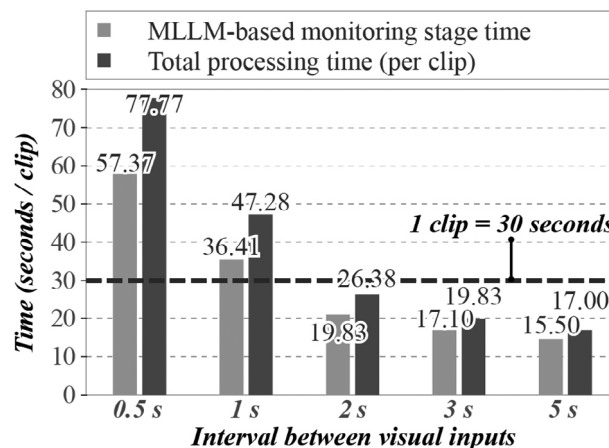


FIGURE 11 Processing time for a single 30-s clip.

ment operations on a minute-by-minute or hourly basis (see Figure 10). This not only reduces the workload for site managers but also enables more precise project documentation, which could enhance the overall quality of project management in the long term. This finding is in line with prior studies, which found that reports generated by LLMs for construction images met pre-established standards for construction inspection report writing, as evaluated by a review team, highlighting the potential for practical application (Pu, Yang, Li, & Guo, 2024).

Figure 11 shows the processing times for a single 30-s clip at different visual input intervals. In the MLLM-based monitoring stage, the API was called twice using



an idle-to-task approach. The results indicate that as the interval between visual inputs increases, the total processing time decreases. Specifically, a 1-s interval resulted in a total processing time of 47.28 s, introducing delays since the ideal processing time should not exceed 30 s. Although the MLLM-based monitoring stage itself required 36.41 s, the zero-shot construction equipment detection stage—if decoupled from the MLLM processing through asynchronous API calls—could be processed in 10.88 s, suggesting that near-real-time performance is attainable. Moreover, at a 2-s interval, the total processing time was 26.38 s, which enables near-real-time monitoring of construction equipment, and also preserves spatial and task inference performance at 95.1% and 90.6%, respectively (see Figure 6). In summary, while processing of a 30-s clip prevents fully real-time monitoring, the proposed framework could be well-suited for applications that require near-real-time feedback such as productivity analysis and equipment operation management. However, its inherent delay may limit its applicability for safety-critical scenarios that demand instantaneous responses.

Regarding price analysis, when the API is called twice per clip for idle-to-task approaches, it costs \$0.014, which amounts to \$3.416 per hour, based on a rate of \$0.15 per 1 million input tokens. By using cached inputs or alternative models (such as Gemini 1.5 Flash), the cost could be reduced to as low as \$1.708 per hour. Although these costs could be significant for sites with heavy equipment usage, they are expected to be manageable for small to medium-sized construction companies operating at sites with lower equipment utilization. Moreover, with the continued downward trend in API pricing for proprietary models, broader adoption of the framework can be anticipated.

Additionally, the proposed framework can be extended to other monitoring tasks, such as worker behavior. However, modifications to the prompt design are necessary; for instance, in Florence-2, the text prompt can be changed to “worker.” Alternatively, a pre-trained deep learning model for human detection could be utilized. Moreover, rather than applying an idle-to-task approach to monitor workers using the MLLM, a prompt design aimed at capturing process-specific information can be adopted. For example, in concrete construction, a prompt could be devised to extract information regarding specific processes such as formwork installation, concrete curing, and compaction. Future research should explore how to effectively adjust the MLLM inputs (i.e., text prompt and visual inputs) for monitoring worker behavior.

5.2 | Limitations

First, the MLLM-based monitoring stage exhibits low accuracy in temporal event questions. This suggests an

inherent challenge for the utilized models in effectively understanding continuous, visually similar image sequences. In contrast, the framework achieves high accuracy in spatial and task inference, making it effective in scenarios where second-by-second temporal resolution is not critical. Future work could fine-tune these models using a dataset consisting of construction site videos and temporal event-related MCQs, which could enable them to better learn the distinctive visual characteristics inherent to construction sites and the nuances of temporal event progression within these environments.

Second, while the proprietary model, GPT-4o mini, was effective in recognizing construction equipment tasks, it has limitations such as API costs. Moreover, some construction companies may not be able to use proprietary models due to security concerns, as their data policy restricts the transmission of sensitive information to external servers. Future research could address these limitations by exploring open-source alternatives.

Third, the current framework only allows the input of a single construction equipment type at a time in the detection stage. To address this limitation, future research could explore a multi-class detection module. Notably, the detection modules in the proposed equipment detection stage can be replaced with alternatives, such as supervised learning models like YOLO, which showed improved performance if sufficient training data are available (see Table 2). While supervised models can enhance detection performance, the proposed ZSL-based detection stage remains effective in addressing the cold-start problem when data are scarce.

Fourth, while the proposed framework demonstrated reliable performance in the case study, it has yet to be extensively validated across diverse construction environments, such as those involving varying lighting conditions, weather scenarios, or equipment types. Broader testing is necessary to ensure its robustness and generalizability under real-world conditions. Furthermore, the framework currently does not consider scenarios involving human-machine coexistence, which is increasingly relevant in modern construction sites. Future work should incorporate such considerations to enhance the framework’s applicability.

Last, it is acknowledged that if equipment is present in later frames but not detected in the first frame, the corresponding clip may be incorrectly skipped. This limitation stems from the current design, which uses the first frame as the key frame to optimize processing efficiency. Additionally, although the SAM2 model is employed for tracking due to its reliability in maintaining object continuity across frames, its performance may degrade in the presence of long-term or dynamic occlusions—common in complex construction environments—potentially affecting the



overall framework's robustness. Future enhancements could include multi-frame scanning to reduce such risks.

6 | CONCLUSION

This study proposed a novel framework for construction equipment monitoring using ZSL and MLLM. In the zero-shot construction equipment detection stage, a detection module using Florence-2 and a tracking module using SAM2 were utilized. In the MLLM-based monitoring stage, two approaches (i.e., idle-to-task approach and cropped visual inputs approach) were introduced to accurately recognize construction equipment tasks. The following key findings and implications are summarized below.

1. The proposed framework achieved an F1-score of 82.2% in detecting construction equipment even in scenarios without data. Moreover, the overall process of the proposed framework is designed to function without training data, using ZSL and MLLM. This eliminates the need for data collection and labeling in construction environments. Its flexibility allows for recognizing construction tasks by modifying text prompts. This could be a particularly advantageous solution for small and medium-sized construction companies lacking resources for extensive data labeling and training.
2. Through the constructed MCQ dataset, the capabilities of the MLLM were systematically examined across three question types—spatial inference, task inference, and temporal event recognition—achieving accuracies of 94.3%, 88.8%, and 59.0%, respectively. This evaluation clarified the scope and limitations of what MLLMs can offer for construction equipment monitoring, while also demonstrating their ability to provide reliable information on equipment location and tasks, contributing to the development of automated monitoring systems.
3. The case study conducted at an actual construction site demonstrated that the proposed system could reliably recognize idle states and operational tasks. This suggests that it can be utilized for automated reporting and productivity analysis in construction projects.

The findings highlight the potential of advanced AI models in facilitating digital transformation within the construction industry, offering a pathway toward efficient and data-driven management practices.

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